

MyGo CVDP 服务器测试报告

DeepSeek V4 Pro -> MyGo -> CIRCT -> Icarus/CVDP cocotb, 2026-06-06

1. 结论摘要

本轮在服务器 Trifoliate 上完整跑完 78 道 CVDP cid003 题目。不带 MyGo 的 direct Verilog baseline 为 37/78 PASS；本次 MyGo 路线为 56/78 PASS，相对 baseline 多通过 19 题。未通过的 22 题中，14 题已经进入 CVDP cocotb 功能判题并失败，8 题停在模型或 MyGo 工具链阶段，未形成有效功能判题结果。

指标	结果	说明
Direct baseline	37/78	DeepSeek 直接生成 Verilog 后交给 CVDP
MyGo server run	56/78	DeepSeek 生成 Go/MyGo，经 MyGo 编译为 HDL 后交给 CVDP
MyGo 净提升	+19 题	只比较最终 PASS 数，不代表每题单调改善
MyGo pass rate	71.79%	本次服务器完整运行结果

2. 状态统计

状态	数量	比例	含义
PASS	56	71.79%	MyGo 产物通过 CVDP 功能判题
FAIL	14	17.95%	HDL 已进入 CVDP cocotb 判题，但功能不符合参考测试
MODEL_OR_MYGO_ERROR	8	10.26%	模型输出或 MyGo/CIRCT 链路未产出可判题 HDL

对照关系	数量	说明
两边都通过	29	Direct 和 MyGo 都 PASS
MyGo 改善	27	Direct FAIL，但 MyGo PASS
MyGo 回退	8	Direct PASS，但 MyGo 未 PASS
两边都失败	14	两条路线都未通过

3. 失败原因解释

这次结果不能简单理解成“CVDP 不能判题”。14 个 FAIL 是 CVDP 已经执行功能测试后的失败，说明当前 MyGo 路线生成的 HDL 行为没有通过 benchmark；8 个 MODEL_OR_MYGO_ERROR 才是模型或 MyGo 工具链层面的失败。

失败类型	数量	解释
CVDP 功能判错	14	生成 HDL 已进入 cocotb 判题，但断言或输出行为不符合参考测试。
MyGo/Go 编译失败	6	模型输出的 Go/MyGo 或 MyGo IR 编译失败。
MyGo/CIRCT Verilog 生成失败	2	MyGo 生成 IR 后在 Verilog/CIRCT 阶段失败。

4. 未通过题目清单

#	题目	Direct	MyGo	类型	原因
3	cvd_p_copilot_64b66b_encoder_0001	PASS	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
7	cvd_p_copilot_apb_dsp_unit_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
8	cvd_p_copilot_apb_gpio_0001	FAIL	MODEL_OR_MYGO_ERROR	MyGo/CIRCT Verilog 生成失败	MyGo compile failed after DeepSeek revision attempts: MyGo verilog FAIL. /home/rongxv/work/mygo_cvd_p_work/08_cvd_p_copilot_apb_gpio_0001_1780710090540_1866677/.mygo-tmp/.mygo-circt-1057250...
9	cvd_p_copilot_apb_history_shift_register_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
24	cvd_p_copilot_concatenate_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
25	cvd_p_copilot_configurable_digital_low_pass_filter_0001	PASS	MODEL_OR_MYGO_ERROR	MyGo/Go 编译失败	MyGo compile failed after DeepSeek revision attempts: MyGo ir FAIL. error: /home/rongxv/work/mygo_cvd_p_work/25_cvd_p_copilot_configurable_digital_low_pass_filter_0001_1780709806034_1866668..
28	cvd_p_copilot_convolutional_encoder_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
30	cvd_p_copilot_data_width_converter_0003	PASS	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
33	cvd_p_copilot_digital_dice_roller_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
35	cvd_p_copilot_edge_detector_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
36	cvd_p_copilot_ethernet_packet_parser_0001	FAIL	MODEL_OR_MYGO_ERROR	MyGo/CIRCT Verilog 生成失败	MyGo compile failed after DeepSeek revision attempts: MyGo verilog FAIL. /home/rongxv/work/mygo_cvd_p_work/36_cvd_p_copilot_ethernet_packet_parser_0001_1780712037989_1866677/.mygo-tmp/.mygo...
39	cvd_p_copilot_fibonacci_series_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
40	cvd_p_copilot_fifo_async_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
43	cvd_p_copilot_gcd_0001	FAIL	MODEL_OR_MYGO_ERROR	MyGo/Go 编译失败	MyGo compile failed after DeepSeek revision attempts: MyGo compile failed
60	cvd_p_copilot_piso_0001	PASS	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
62	cvd_p_copilot_restoring_division_0001	FAIL	MODEL_OR_MYGO_ERROR	MyGo/Go 编译失败	MyGo compile failed after DeepSeek revision attempts: MyGo ir FAIL. error: /home/rongxv/work/mygo_cvd_p_work/62_cvd_p_copilot_restoring_division_0001_1780712342790_1866671/main.go:64:1: mis...
64	cvd_p_copilot_secure_read_write_bus_0001	PASS	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
65	cvd_p_copilot_secure_read_write_register_bank_0001	FAIL	MODEL_OR_MYGO_ERROR	MyGo/Go 编译失败	MyGo compile failed after DeepSeek revision attempts: MyGo compile failed
68	cvd_p_copilot_set_bit_calculator_0001	PASS	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
70	cvd_p_copilot_static_branch_predict_0001	FAIL	FAIL	CVDP 功能判错	CVDP cocotb reported failing tests despite zero process return code.
75	cvd_p_copilot_unpacker_one_hot_0001	PASS	MODEL_OR_MYGO_ERROR	MyGo/Go 编译失败	MyGo compile failed after DeepSeek revision attempts: MyGo compile failed
77	cvd_p_copilot_vga_controller_0001	PASS	MODEL_OR_MYGO_ERROR	MyGo/Go 编译失败	MyGo compile failed after DeepSeek revision attempts: MyGo ir FAIL. error: /home/rongxv/work/mygo_cvd_p_work/77_cvd_p_copilot_vga_controller_0001_1780714849469_1866668/main.go:95:1: missing...

5. 完整 78 题对照表

#	题目	难度	Direct	MyGo	对照	MyGo 说明
1	cvdp_copilot_16qam_mapper_0001	medium	FAIL	PASS	MyGo 改善	通过
2	cvdp_copilot_16qam_mapper_0006	medium	FAIL	PASS	MyGo 改善	通过
3	cvdp_copilot_64b66b_encoder_0001	easy	PASS	FAIL	MyGo 回退	CVDP cocotb reported failing tests despite zero process return code.
4	cvdp_copilot_8x3_priority_encoder_0001	easy	PASS	PASS	两边都通过	通过
5	cvdp_copilot_Carry_Lookahead_Adder_0001	easy	PASS	PASS	两边都通过	通过
6	cvdp_copilot_GFCM_0001	easy	FAIL	PASS	MyGo 改善	通过
7	cvdp_copilot_apb_dsp_unit_0001	medium	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.
8	cvdp_copilot_apb_gpio_0001	medium	FAIL	MODEL_OR_MYGO_ERROR	两边都失败	MyGo compile failed after DeepSeek revision attempts: MyGo verilog FAIL. /home/rongxv/work/mygo_cvdp_work/08_cvdp_cop...
9	cvdp_copilot_apb_history_shift_register_0001	medium	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.
10	cvdp_copilot_axi_register_0001	medium	FAIL	PASS	MyGo 改善	通过
11	cvdp_copilot_axi_stream_upscale_0001	easy	FAIL	PASS	MyGo 改善	通过
12	cvdp_copilot_axil_precision_counter_0001	medium	FAIL	PASS	MyGo 改善	通过
13	cvdp_copilot_axis_joiner_0001	medium	PASS	PASS	两边都通过	通过
14	cvdp_copilot_barrel_shifter_0001	easy	PASS	PASS	两边都通过	通过
15	cvdp_copilot_bcd_counter_0001	easy	PASS	PASS	两边都通过	通过
16	cvdp_copilot_bcd_to_excess_3_0001	easy	PASS	PASS	两边都通过	通过
17	cvdp_copilot_binary_to_one_hot_decoder_0001	easy	PASS	PASS	两边都通过	通过
18	cvdp_copilot_caesar_cipher_0001	easy	PASS	PASS	两边都通过	通过
19	cvdp_copilot_car_parking_management_0001	medium	PASS	PASS	两边都通过	通过
20	cvdp_copilot_cascaded_adder_0001	easy	PASS	PASS	两边都通过	通过
21	cvdp_copilot_clock_divider_0003	easy	PASS	PASS	两边都通过	通过
22	cvdp_copilot_comparator_0001	easy	PASS	PASS	两边都通过	通过
23	cvdp_copilot_complex_multiplier_0001	easy	PASS	PASS	两边都通过	通过
24	cvdp_copilot_concatenate_0001	medium	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.
25	cvdp_copilot_configurable_digital_low_pass_filter_0001	medium	PASS	MODEL_OR_MYGO_ERROR	MyGo 回退	MyGo compile failed after DeepSeek revision attempts: MyGo ir FAIL. error: /home/rongxv/work/mygo_cvdp_work/25_cvdp_c...
26	cvdp_copilot_configurable_digital_low_pass_filter_0004	medium	PASS	PASS	两边都通过	通过
27	cvdp_copilot_configurable_digital_low_pass_filter_0004	easy	FAIL	PASS	MyGo 改善	通过
28	cvdp_copilot_convolutional_encoder_0001	easy	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.
29	cvdp_copilot_data_bus_controller_0001	medium	FAIL	PASS	MyGo 改善	通过
30	cvdp_copilot_data_width_converter_0003	easy	PASS	FAIL	MyGo 回退	CVDP cocotb reported failing tests despite zero process return code.
31	cvdp_copilot_dbi_0001	medium	PASS	PASS	两边都通过	通过
32	cvdp_copilot_decode_firstbit_0001	medium	FAIL	PASS	MyGo 改善	通过
33	cvdp_copilot_digital_dice_roller_0001	easy	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.

#	题目	难度	Direct	MyGo	对照	MyGo 说明
34	cvdp_copilot_digital_stopwatch_0001	easy	FAIL	PASS	MyGo 改善	通过
35	cvdp_copilot_edge_detector_0001	easy	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.
36	cvdp_copilot_ethernet_packet_parser_0001	medium	FAIL	MODEL_OR_MYGO_ERROR	两边都失败	MyGo compile failed after DeepSeek revision attempts: MyGo verilog FAIL. /home/rongxv/work/mygo_cvdp_work/36_cvdp_cop...
37	cvdp_copilot_events_to_apb_0001	easy	FAIL	PASS	MyGo 改善	通过
38	cvdp_copilot_factorial_0001	medium	PASS	PASS	两边都通过	通过
39	cvdp_copilot_fibonacci_series_0001	easy	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.
40	cvdp_copilot_fifo_async_0001	medium	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.
41	cvdp_copilot_filo_0005	medium	PASS	PASS	两边都通过	通过
42	cvdp_copilot_fsm_seq_detector_0001	easy	PASS	PASS	两边都通过	通过
43	cvdp_copilot_gcd_0001	easy	FAIL	MODEL_OR_MYGO_ERROR	两边都失败	MyGo compile failed after DeepSeek revision attempts: MyGo compile failed
44	cvdp_copilot_gf_multiplier_0001	easy	PASS	PASS	两边都通过	通过
45	cvdp_copilot_hamming_code_tx_and_rx_0001	easy	PASS	PASS	两边都通过	通过
46	cvdp_copilot_hamming_code_tx_and_rx_0003	easy	PASS	PASS	两边都通过	通过
47	cvdp_copilot_hebbian_rule_0017	medium	FAIL	PASS	MyGo 改善	通过
48	cvdp_copilot_hill_cipher_0001	medium	FAIL	PASS	MyGo 改善	通过
49	cvdp_copilot_load_store_unit_0001	medium	PASS	PASS	两边都通过	通过
50	cvdp_copilot_matrix_multiplier_0001	easy	PASS	PASS	两边都通过	通过
51	cvdp_copilot_microcode_sequencer_0001	medium	FAIL	PASS	MyGo 改善	通过
52	cvdp_copilot_morse_code_0001	easy	PASS	PASS	两边都通过	通过
53	cvdp_copilot_moving_average_0001	easy	PASS	PASS	两边都通过	通过
54	cvdp_copilot_nbit_swizzling_0001	easy	FAIL	PASS	MyGo 改善	通过
55	cvdp_copilot_packet_controller_0001	medium	FAIL	PASS	MyGo 改善	通过
56	cvdp_copilot_palindrome_3b_0002	easy	PASS	PASS	两边都通过	通过
57	cvdp_copilot_perceptron_0001	medium	FAIL	PASS	MyGo 改善	通过
58	cvdp_copilot_perf_counters_0001	easy	FAIL	PASS	MyGo 改善	通过
59	cvdp_copilot_perfect_squares_0001	easy	FAIL	PASS	MyGo 改善	通过
60	cvdp_copilot_piso_0001	easy	PASS	FAIL	MyGo 回退	CVDP cocotb reported failing tests despite zero process return code.
61	cvdp_copilot_prbs_gen_0003	medium	FAIL	PASS	MyGo 改善	通过
62	cvdp_copilot_restoring_division_0001	medium	FAIL	MODEL_OR_MYGO_ERROR	两边都失败	MyGo compile failed after DeepSeek revision attempts: MyGo ir FAIL. error: /home/rongxv/work/mygo_cvdp_work/62_cvdp_c...
63	cvdp_copilot_reverse_bits_0001	easy	PASS	PASS	两边都通过	通过
64	cvdp_copilot_secure_read_write_bus_0001	medium	PASS	FAIL	MyGo 回退	CVDP cocotb reported failing tests despite zero process return code.
65	cvdp_copilot_secure_read_write_register_bank_0001	medium	FAIL	MODEL_OR_MYGO_ERROR	两边都失败	MyGo compile failed after DeepSeek revision attempts: MyGo compile failed
66	cvdp_copilot_sequential_binary_to_one_hot_decoder_0001	easy	FAIL	PASS	MyGo 改善	通过
67	cvdp_copilot_serial_in_parallel_out_0004	easy	PASS	PASS	两边都通过	通过
68	cvdp_copilot_set_bit_calculator_0001	easy	PASS	FAIL	MyGo 回退	CVDP cocotb reported failing tests despite zero process return code.
69	cvdp_copilot_sorter_0001	easy	FAIL	PASS	MyGo 改善	通过

#	题目	难度	Direct	MyGo	对照	MyGo 说明
70	cvdp_copilot_static_branch_predict_0001	medium	FAIL	FAIL	两边都失败	CVDP cocotb reported failing tests despite zero process return code.
71	cvdp_copilot_sync_lifo_0001	medium	FAIL	PASS	MyGo 改善	通过
72	cvdp_copilot_sync_serial_communication_0001	medium	FAIL	PASS	MyGo 改善	通过
73	cvdp_copilot_thermostat_0001	medium	PASS	PASS	两边都通过	通过
74	cvdp_copilot_ttc_lite_0001	medium	FAIL	PASS	MyGo 改善	通过
75	cvdp_copilot_unpacker_one_hot_0001	easy	PASS	MODEL_OR_MYGO_ERROR	MyGo 回退	MyGo compile failed after DeepSeek revision attempts: MyGo compile failed
76	cvdp_copilot_vending_machine_0001	medium	FAIL	PASS	MyGo 改善	通过
77	cvdp_copilot_vga_controller_0001	medium	PASS	MODEL_OR_MYGO_ERROR	MyGo 回退	MyGo compile failed after DeepSeek revision attempts: MyGo ir FAIL. error: /home/rongxv/work/mygo_cvdv_work/77_cvdv_c...
78	cvdp_copilot_wb2ahb_0001	medium	FAIL	PASS	MyGo 改善	通过

数据文件

服务器汇总: cvdp-results\deepseek-v4-mygo-server-20260606\server_merged_summary.json
对照明细: cvdp-results\deepseek-v4-mygo-server-20260606\server_vs_direct_detailed.csv
Direct baseline: tests\verilog-eval\historical\runs\cvdp_cid003_deepseek_v4_direct_20260602
运行设置: 4 shards, skip-existing enabled, go-attempts=3, retries=2; MyGo commit 125084e